Early Disease Detection — Technical Documentation

Oct 23, 2026 · @Syed Jafri

Overview

The project trains a classifier on repeat multispectral and thermal imagery, labelled by ground-truthed plots, to flag infected plants or plot zones *before* symptoms are visible to a scout — and measures how many days of lead time that actually buys.

Why it can work — infection often changes plant physiology before visible lesions: stomatal closure raises leaf temperature (thermal), and early chlorophyll and cell-structure changes shift red-edge and NIR reflectance. The project tests how early these signals separate infected from healthy plots for the target pathosystem.

Scope

- In: repeat drone flights (multispectral and/or thermal), ground-truth plots with dated disease assessments and, ideally, lab confirmation; weather data.
- Out: per-plot and per-pixel infection probability maps per flight, lead-time estimates, alert zones for scouting, and a validation report.
- Not in scope: identifying the pathogen species from imagery alone, treatment recommendations, and satellite-scale monitoring (v1).

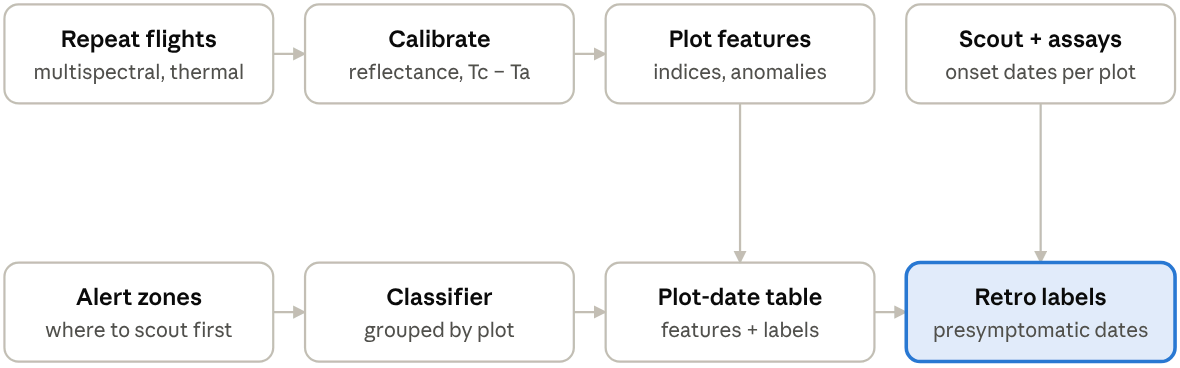
Key definitions

Term	Meaning
Pre-symptomatic	Plot is infected (confirmed later or by assay) but a trained scout sees no symptoms on that date
Onset date	First date visible symptoms are recorded in the plot
Lead time	Onset date − first date the model flags the plot above threshold

Pipeline

Two streams meet in a plot-date table: imagery features for every flight, and labels assigned backwards in time from when symptoms or assays later confirm infection.

Later scout assessments label earlier, healthy-looking flights



pipeline · imagery and ground truth to alerts

Stack: Python 3.11, rasterio/GDAL, geopandas, xarray for the flight stack, scikit-learn and XGBoost, PyTorch for temporal and patch models, and a Typer CLI `edd`.

Imagery: multispectral and thermal

Pre-symptomatic signals are small, so radiometric consistency between flights matters more than spatial resolution; thermal data in particular is only usable when normalized for weather at flight time.

Acquisition

Parameter	Multispectral	Thermal
Bands	Blue, Green, Red, Red Edge, NIR	LWIR 7.5–13.5 µm, radiometric
Ground sample distance	2–5 cm	5–15 cm (lower native resolution)
Calibration	Reflectance panel pre/post flight + downwelling light sensor	Ground reference targets of known temperature (e.g. water tank, black/white panels) + blackbody check

Parameter	Multispectral	Thermal
Timing	Within ±2 h of solar noon	Midday, clear sky, stable wind (< 4 m/s); stress signal strongest when transpiration demand is high
Revisit	Every 3–7 days during the infection risk window	Same flights
Weather log	—	Air temperature, relative humidity, wind, solar radiation at flight time

Preprocessing

1. **Radiometric calibration** — empirical-line reflectance for multispectral; emissivity correction (≈ 0.98 for leaves) and target-based offset for thermal.
2. **Co-registration** — all flights warped to a season reference grid; thermal is registered to the multispectral orthomosaic (sub-pixel RMSE target ⩽ 1 thermal pixel).
3. **Canopy masking** — NDVI threshold plus thermal mixing check; only sunlit canopy pixels are kept, since soil and shadow dominate thermal readings.
4. **Thermal normalization** — compute canopy-minus-air temperature (Tc − Ta) and a crop water stress index (CWSI) using wet and dry reference temperatures per flight; raw temperature is never compared across dates.
5. **Plot extraction** — zonal statistics inside each plot polygon, shrunk by an edge buffer to avoid border effects.

Ground-truth plots and labels

The hardest part of pre-symptomatic detection is knowing a plot was infected on a date when it looked healthy; labels are therefore assigned retrospectively from later assessments and, where possible, lab assays.

Plot design

- Fixed, georeferenced plots (e.g. 1–3 m² or a set number of plants), marked with ground targets visible in imagery.
- Include healthy controls, naturally exposed plots and, where the study allows, inoculated plots with known inoculation dates, which give the cleanest timeline.

- Randomized block layout so infection status is not confounded with field position, irrigation or soil.
- Abiotic stress controls (e.g. water-stressed but uninfected plots) to check the model detects disease, not stress in general.

Assessments — on every flight date (or within 24 h): visual severity score (e.g. 0–5 or % leaf area affected) by a trained scout blind to model output, photos, and, on a subset, tissue samples for qPCR, ELISA or culture confirmation.

Label assignment per plot-date

Label	Rule
`healthy`	Never symptomatic through season end and assay-negative where tested
`presymptomatic`	Visibly healthy on this date, but symptomatic later (or assay-positive now)
`symptomatic`	Visible symptoms recorded on this date
`uncertain`	Abiotic damage, missed assessment, or conflicting assay; excluded from training

A `presymptomatic` label is only used if onset occurs within `labels.max_lead_days` (default 14) after the date, so very early dates are not labelled infected on weak evidence. The days-to-onset value is kept as a continuous target for regression and lead-time analysis.

Sample size — aim for ⩾ 30 infected and ⩾ 30 healthy plots per season and site; power depends on effect size, so a pilot season to estimate it is recommended.

Features

Each plot-date gets spectral, thermal, texture and temporal features; temporal change relative to the plot's own baseline and to nearby healthy plots usually carries more signal than any single-date value.

Spectral indices

Index	Formula	Sensitive to
NDVI	(NIR − R) / (NIR + R)	Biomass, canopy cover (late signal)
NDRE	(NIR − RE) / (NIR + RE)	Chlorophyll in dense canopy
CIre (red-edge chlorophyll index)	NIR / RE − 1	Chlorophyll content
GNDVI	(NIR − G) / (NIR + G)	Chlorophyll
PRI*	(R531 − R570) / (R531 + R570)	Photosynthetic efficiency; needs narrow bands

*Only with a sensor that has these bands.

Thermal — Tc − Ta, CWSI, within-plot temperature standard deviation and 90th percentile (infection often starts patchy).

Within-plot statistics — mean, median, SD, 10th/90th percentiles and coefficient of variation of each index; GLCM texture (contrast, homogeneity) on NDRE.

Temporal features

- Change since previous flight (Δ) and since the plot's baseline (mean of first two flights).
- Anomaly vs reference: z-score of the plot against healthy-control plots on the same date, which removes weather and phenology effects shared by all plots.
- Slope of each index over the last 3 flights.

Covariates — growth stage, days after planting, cumulative leaf-wetness hours or disease-risk index from weather (as context, not as a substitute for imagery), and block ID for mixed models.

Pixel-level variant — for mapping, the same indices are computed per pixel (or per 0.5 m cell) with temporal anomaly features, feeding a pixel classifier or a CNN on image patches.

Models, training and validation

The primary task is binary — `presymptomatic` vs `healthy` on dates when both look the same to a scout — because that is the claim the project makes; symptomatic plots are reported separately so they cannot inflate accuracy.

Models

Model	Input	Role
Logistic regression (regularized)	Plot features	Baseline, interpretable coefficients
Random forest / XGBoost	Plot features	Main tabular model
Temporal model (1D CNN or GRU)	Sequence of plot feature vectors up to the date	Uses trajectory shape; only if ⩾ ~200 plot sequences
Patch CNN (e.g. ResNet-18, 5–6 input channels)	Image patches + temporal anomaly channels	Pixel-level maps; needs more labels

Class handling — infected plots are usually a minority; use class weights and report PR-AUC, not accuracy.

Validation — preventing leakage

- **Group by plot**: all dates from one plot are in the same fold, so the model never sees a plot's later (symptomatic) dates when tested on its early ones.
- **Group by block or field** for spatial independence; **leave-one-season-out** or **leave-one-site-out** for the headline generalization result.
- **Forward-in-time rule**: features for date *t* use only flights up to *t*; no smoothing across future dates.
- Feature selection, tuning (nested CV) and normalization fit inside training folds only.

Metrics

- Presymptomatic vs healthy: ROC-AUC, PR-AUC, sensitivity at 90% specificity.
- Performance by days-to-onset bin (1–3, 4–7, 8–14 days before onset).
- Specificity on abiotic-stress controls (false alarm rate for drought etc.).
- Calibration (reliability curve, Brier score).

Explanation — permutation importance and SHAP grouped by feature family (thermal, red-edge, temporal) to report which signals appear first, checked for stability across folds.

Lead time and alerting

The practical question is how many days earlier than a scout the model raises a reliable flag, at a false-alarm rate a grower would tolerate.

Lead time per infected plot — run the model forward through the season using only data available on each date. The first alert date is the first flight where probability ≥ threshold *and* stays above it on the next flight (two-flight confirmation reduces one-off false alarms). Lead time = onset date − first alert date; plots never flagged before onset get lead time 0 (missed).

Reported results

- Distribution of lead times (median, IQR) for infected plots.
- Share of infected plots flagged ≥ 3 and ≥ 7 days before onset.
- False alerts per healthy plot per season, and on abiotic-stress controls.
- A trade-off curve: median lead time vs false-alert rate across thresholds.

Operational alerting

1. After each flight, score every plot or 0.5 m grid cell.
2. Cluster cells above threshold into alert zones (min area `alerts.min_area_m2`), ranked by probability × area.
3. Export zones with centroid coordinates for targeted scouting, plus an image chip per zone.
4. Scout feedback (confirmed / not confirmed / abiotic) is logged and fed back as new labels.

Alerts are framed as *where to scout first*, not as a diagnosis.

Configuration, CLI, limitations and roadmap

One config per season and pathosystem drives the pipeline; every stage writes to `derived/` and can be rerun alone.

`config/season.yaml`

```
season: 2026_potato_lateblight
pathosystem: potato / Phytophthora infestans   # example; set yours
plots: data/plots.gpkg
assessments: data/assessments.csv      # plot_id, date, severity, assay
weather: data/weather.csv
flights: data/flights/{date}/{ms|th}.tif
reference_plots: [C01, C02, C03, C04]  # healthy controls for anomalies
labels:
  max_lead_days: 14
features:
  indices: [NDVI, NDRE, CIre, GNDVI]
  thermal: [tc_minus_ta, cwsi, t_sd, t_p90]
  edge_buffer_m: 0.2
model:
  type: xgboost                         # logreg | rf | xgboost | gru | cnn
  cv: {group: plot, outer: leave_one_season_out}
alerts:
  threshold: auto                       # chosen at 90% specificity
  confirm_flights: 2
  min_area_m2: 1
```

CLI

```
edd preprocess --config config/season.yaml --date 2026-07-10
edd features   --config config/season.yaml
edd labels     --config config/season.yaml
edd train      --config config/season.yaml
edd leadtime   --config config/season.yaml          # forward-in-time replay
edd alert      --config config/season.yaml --date 2026-07-10
```

Limitations

- Signals are pathosystem-specific; a model trained on one crop–disease pair should not be assumed to transfer.
- Thermal and many spectral changes are not disease-specific: drought, nutrient deficiency and herbicide injury can look similar; abiotic controls quantify, but do not remove, this.
- Broadband multispectral sensors lack narrow bands (e.g. for PRI) that some studies rely on.
- Retrospective labels depend on assessment quality; inoculated plots give cleaner timelines than natural infection but may not represent field conditions.

- Lead time is bounded by revisit interval; 7-day flights cannot show 2-day lead times.

Roadmap

- [ ] Hyperspectral sensor trial for narrow-band indices
- [ ] Combine weather-based infection risk models as a prior
- [ ] Multi-season, multi-site training set
- [ ] Scout feedback app writing labels straight into the training data